4.1.2 Atomic structure

The study of atomic structure provides a good opportunity to show how scientific methods and theories develop over time. The model introduced in this topic describes atoms in terms of a central nucleus with protons and neutrons surrounded by electrons in a series of energy levels (shells). The ideas in this topic can account for the existence of isotopes and underpin the study of radioactivity (Radiation and risk (page 55)), chemical bonding (Structure and bonding (page 98)) and the periodic table (The periodic table (page 87)).

4.1.2.1 Scientific models of the atom

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how and why the atomic model has changed over time.	Stages in the development of atomic models: • Dalton atoms (1804) – spherical atoms that cannot be split up to explain the properties of gases and the formulae of compounds • Plum pudding model (1897) – it was found that the mass of electrons, which had recently been discovered, was very much less than the mass of atoms so they must be sub-atomic particles • the nuclear atom (1911) – an experiment which showed that most of the alpha particles directed at thin gold foil passed through but a few bounced back, suggesting the positive charge was concentrated at the centre of each gold atom • discovery of neutrons in the nucleus (1932) – explained why the mass of atoms was greater than could be accounted for by the mass of the protons. Students are not required to recall dates or the names of scientists.	WS 1.1 Explain, with examples, why new data from experiments or observations led to changes in atomic models. Decide whether or not given data supports a particular theory.

4.1.2.2 The size of atoms

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the typical size (order of magnitude) of atoms and small molecules.	Atoms are very small, having a radius of about 0.1 nm (1×10^{-10} m). The radius of a small molecule such as methane, CH_4, is about 0.5 nm (5×10^{-10} m).	MS 1b Interpret expressions in standard form. WS 4.4 Use SI units and the prefix nano. MS 1d Estimate the size of atoms based on scale diagrams.

4.1.2.3 Sub-atomic particles

<table>
<tr><th>GCSE science subject content</th><th>Details of the science content</th><th>Scientific, practical and mathematical skills</th></tr>
<tr><td>Describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with almost all of the mass in the nucleus.

Recall that atomic nuclei are composed of both protons and neutrons, that the nucleus of each element has a characteristic positive charge, but that elements can differ in nuclear mass by having different numbers of neutrons.

Recall relative charges and approximate relative masses of protons, neutrons and electrons.</td><td>The radius of a nucleus is less than 1/10 000 of that of the atom (about 1 x 10^{-14} m).

The relative masses and charges of protons, neutrons and electrons are:

<table><tr><th>Name of particle</th><th>Relative mass</th><th>Charge</th></tr><tr><td>Proton</td><td>1</td><td>+1</td></tr><tr><td>Neutron</td><td>1</td><td>0</td></tr><tr><td>Electron</td><td>Very small</td><td>–1</td></tr></table>
The number of protons in an atom of an element is its atomic number. All atoms of a particular element have the same number of protons. Atoms of different elements have different numbers of protons.

In an atom, the number of electrons is equal to the number of protons in the nucleus. Atoms have no overall electrical charge.</td><td>WS 1.2

Interpret and draw diagrams to represent atoms.</td></tr>
</table>

4.1.2.4 Isotopes

<table>
<tr><th>GCSE science subject content</th><th>Details of the science content</th><th>Scientific, practical and mathematical skills</th></tr>
<tr><td>Relate differences between isotopes to differences in conventional representations of their identities, charges and masses.</td><td>The sum of the protons and neutrons in an atom is its mass number.

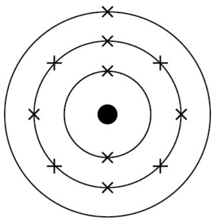
Atoms of the same element can have different numbers of neutrons; these atoms are called isotopes of that element.

Atoms can be represented as shown in this example:

(Mass number) 23
(Atomic number) 11 $^{23}_{11}\text{Na}$</td><td>WS 1.2

Work out numbers of protons, neutrons and electrons in atoms and ions, given atomic number and mass number of isotopes.</td></tr>
</table>

4.1.2.5 Electrons in atoms

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that in each atom its electrons are arranged at different distances from the nucleus.	The electrons in an atom occupy the lowest available energy levels (innermost available shells closest to the nucleus). The electronic structure of an atom can be represented by numbers or by a diagram. For example, the electronic structure of sodium is 2,8,1 or  showing two electrons in the lowest energy level, eight in the second energy level and one in the third energy level.	This topic links with Atomic number and the periodic table (page 87).